Computer Networks 2 - Exam March 2024

Maximum points: 36p
Preliminary grading: Pass (3): 19p, (4): 26p, (5): 32p

NAME / ANONYMOUS NUMBER:

All answers must be in English. Start a new page for each problem.

Read the questions carefully and answer what is asked for! Your task is to convince the teachers who are grading the exam that you have a certain level of knowledge. The best way to do this are concise, comprehensible and clear answers to the questions asked – and nothing else.
In some of the problems, you may not be given all the details needed to answer the problem. In such cases, you need to make realistic assumptions that are explicitly stated and motivated. Omitting an assumption or not motivating your assumptions will give you less points.

Also, remember to **write in a clear, readable handwriting**.

Good luck with the exam!

Problem Number	Solved	Points
1		
2		
3		
4		
5		
6		
Sum		
Bonus Points		
Grade		

Please set a X in the Solved column for each problem you have worked on.

Problem 1: Short Questions (10p)

In this problem you can give short answers without motivation, unless stated differently.

a. You connect to a new server using QUIC. Can your QUIC client perform a zero-RTT connection establishment to that server?

b. Describe briefly an example of a scenario where a connection migration of a QUIC connection is needed.

c. Assume a random variable X that follows a geometric distribution with p=0.4. Which value is higher: P(X=3) or P(X=4)?

d. The binomial distribution is a discrete probability distribution of the number of successes of a series of independent Bernoulli experiments. Is this correct?

e. Mention one characteristic of the Kubernetes networking model.

f. What is the main benefit of passive network measurements over active network measurements?

g. Mention two problems with the host-centric view of today's Internet (max one sentence per problem).

h. What is the difference between transmission delay and queueing delay?

i. What is one of the main performance bottlenecks of today's networks?

j. Give three examples of network performance metrics.

Problem 2: Power-saving (3p)

Given the increase in cost for power, servers are typically only on when being used, and turned off whenever they are not in use. Assume now the following power-aware algorithm: when a job arrives, a fresh server is turned on (assume zero setup cost). As soon as the job completes service, the server is turned off. Moreover, assume that there is always a server available for every job, so there is no queueing. Assume that when a server is on, it consumes power at a rate of $P = 120$ Watts. Assume $\lambda = 20$ jobs arrive per second, and that the service requirement of jobs is exponentially-distributed with mean 10 seconds.

Derive the average rate at which power is used in the system.

Problem 3: Queueing (7p)

Assume an M/M/K/2K server system, i.e., a networking system with K servers and some queueing capacity. The arrival rate is λ, the service rate of each server is μ, and the packet loss due to full buffer is P_{block}. For this system, we can define the throughput as $\lambda(1 - P_{block})$.

Answer the following questions, and justify each answer (1-5 lines max for each question). Note that you do not need to perform any derivations using the formulas (unless you prefer).

a) What is the queueing capacity of the system? (1p)
b) Assume that you increase the number of servers to 2K, while everything else stays the same, including the queueing capacity. Does the throughput of the server system increase or decrease or stay the same? (2p)
c) Continue with the assumption from question b. Does the service time E[S] for the jobs that receive service increase, decrease or stay the same? (1p)
d) Assume now that you keep the servers to K, but you duplicate the queueing capacity. Does the throughput of the server increase or decrease or stay the same? (1p)
e) Continue with the assumption from question d. Does the service time E[S] for the jobs that receive service increase, decrease or stay the same? (1p)
f) Continue with the assumption from question d. Does the mean response time E[W] for the jobs that receive service increase, decrease or stay the same? (1p)

Problem 4: Markov Chains (7p)

Consider a network router that operates in three states: normal, congested, and failed.
In the normal state:

- It remains in the normal state with a probability of 0.9.
- It transitions to the congested state with a probability of 0.05.
- It transitions to the failed state with a probability of 0.05.

In the congested state:

- It stays congested with a probability of 0.5.
- It returns to the normal state with a probability of 0.1.
- It transitions to the failed state with a probability of 0.4.

In the failed state:

- It remains failed with a probability of 1.
- It transitions to the normal state with a probability of 0.
- It transitions to the congested state with a probability of 0.

Now, answer the following questions:

a) Derive the state transition matrix and draw the Markov Chain. (1p)
b) Is the Markov Chain irreducible? Motivate your answer. If not, explain what change(s) are necessary to make it irreducible. (2p)
c) A network operator reconfigures the router and the probabilities in the failed state change. Now, the network remains failed with a probability of 0.2, transitions to the normal state with probability 0.6, or transitions to the congested state with probability 0.2. Derive the new state transition matrix and draw the new Markov Chain. (1p)
d) Assume the new network router configuration starts in the normal state. Calculate the probability vector after three-time units. (2p)
e) Write a pseudocode to compute the stationary distribution of the new Markov Chain. (1p)

Problem 5: Channel Capacity (3p)

Consider the Channel Capacity Theorem for a continuous communication channel with bandwidth W Hz, perturbed by the constant additive white Gaussian noise of power spectral density N_0, and the average transmitted power P.

Answer the following questions:

a) Is there any limit to the capacity of such a channel if you increase its signal-to-noise ratio $\frac{P}{N_0 W}$ without limit? If so, what is that limit? (1p)

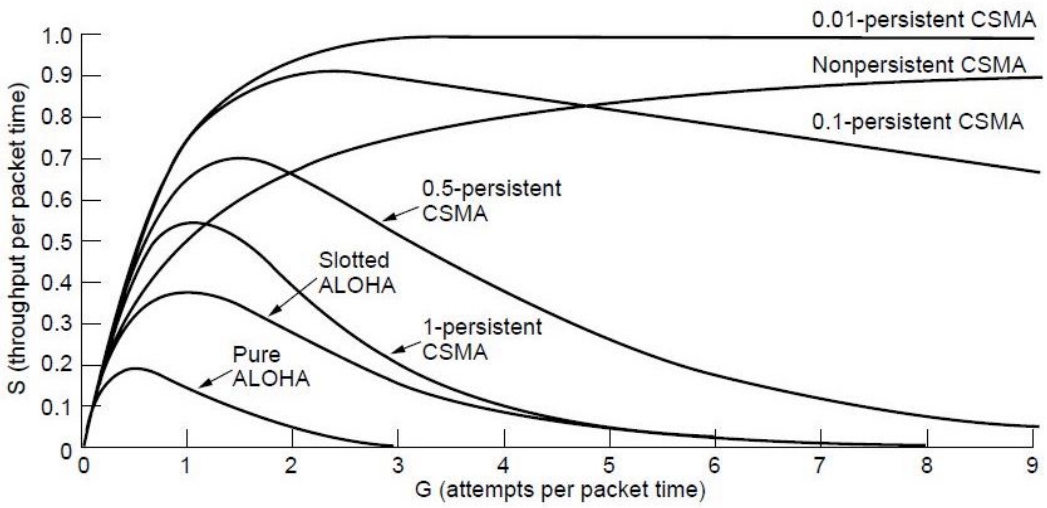
b) Consider the situation in question a. What are the impacts of such increase in the signal-to-noise ratio without limits? Motivate your answer. (1p)

c) If you can increase the bandwidth W without limits, while fixing P, how will the Channel Capacity be affected? (1p)

Problem 6: ALOHA and CSMA (6p)

One important difference between ALOHA and CSMA is carrier sensing, which means that CSMA listens for ongoing transmissions to terminate before starting a transmission. Transmissions after the transmission medium becomes idle can be done more or less "aggressive" based on a parameter p (p-persistent CSMA) as follows:

> "When the transmitting node is ready to transmit data, it senses the transmission medium for idle or busy. If idle, then it transmits immediately. If busy, then it senses the transmission medium continuously until it becomes idle, then transmits with probability p. If the node does not transmit (the probability of this event is 1-p), it waits until the next available time slot. If the transmission medium is not busy, it transmits again with the same probability p. This probabilistic hold-off repeats until the frame is finally transmitted or when the medium is found to become busy again (i.e. some other node has already started transmitting). In the latter case the node repeats the whole logic cycle (which started with sensing the transmission medium for idle or busy) again."

The following figure compares ALOHA and CSMA. For this problem, you can consider S on the y-axis to be the efficiency of the protocols, and you should ignore G on the x-axis. In other words, consider the maximum efficiency of each protocol.

a. Relating to probability theory, argue why Slotted ALOHA achieves higher efficiency than Pure ALOHA (nodes can start to transmit at any time). That is, for both protocols explain when a collision occurs and relate the probabilities to each other. You do not have to calculate the maximum efficiency. (2p)
b. Argue why 0.01-persistent CSMA achieves higher efficiency than 1-persistent CSMA. (2p)
c. What is the drawback of CSMA with lower persistency p? Discuss the trade-off between efficiency and this drawback. (2p)